

ICH E6 - GCP

1. Definition:

The International Council for Harmonisation (ICH) has long provided global guidance on the design, conduct, recording, and reporting of clinical trials. The E6(R3) guideline represents the latest revision of Good Clinical Practice (GCP), consolidating the earlier framework into 11 principles.

These principles reflect decades of experience, the ethical foundation of the [Declaration of Helsinki](#), and modern regulatory expectations. They place participants' rights, safety, and well-being at the centre of clinical research, while ensuring that trials are scientifically sound, operationally feasible, proportionate to risk, and based on quality by design.

For professionals in biotech, pharma, [CROs](#), and clinical sites, these principles are not abstract ethics — they are daily operational requirements that shape protocol design, patient recruitment, monitoring, data integrity, and regulatory compliance.

2. The 11 Principles of ICH Good Clinical Practice E6(R3)

2.1 Principle 1

“Clinical Trial should be conducted in accordance with the ethical principles that have their origin in the Declaration of Helsinki and that are consistent with GCP and applicable regulatory requirement(s). Clinical trials should be designed and conducted in ways that ensure the rights, safety and well-being of participants.”

What it means:

Ethical principles guide every aspect of clinical research. Subpoints (1.1–1.6) explain that:

- Participants' rights and safety always come before scientific or societal interests.
- Ongoing safety reviews are mandatory as new data emerges.
- Risks must be justified by expected benefits.
- Populations should not be unnecessarily excluded, promoting diversity and generalizability.
- Medical care must be overseen by qualified physicians or other professionals.
- Participant confidentiality must be protected.

Impact on trials:

- Trial initiation depends on a justified risk–benefit balance.
- Study designs must consider diversity and avoid unjustified exclusions.
- Ethical oversight continues throughout the trial, not just at start-up.
- Confidentiality and privacy regulations (e.g., GDPR) must be strictly observed.

Who should pay attention in Principle 1 of the 11 Principles of ICH:

- **Sponsors:** Ensure ethical trial design, appropriate inclusion/exclusion criteria, and continuous safety evaluation.
- **CROs:** Oversee compliance with ethics during monitoring and ensure adverse events are reported.
- **Sites/SMOs:** Safeguard participants daily, obtain consent properly, and provide medical oversight.
- **Labs:** Maintain confidentiality of patient data and ensure sample handling respects privacy and safety.

2.2 Principle 2

“Informed Consent is an integral feature of the ethical conduct of a trial. Clinical Trial participation should be voluntary and based on a consent process that ensures participants (or their legally acceptable representatives, where applicable) are well-informed following the regulatory requirements and the processes approved by the IRB/IEC.”

What it means:

Informed consent is not just a form; it is a process. Subpoints (2.1–2.4) highlight that:

- Consent must be obtained freely and documented before participation.
- Participants or their legally acceptable representatives must clearly understand risks, benefits, and burdens.
- The process should adapt to trial design, participant characteristics, and context — including emergency or technology-assisted consent.
- In emergencies, delayed consent must follow strict regulations and ethics approval.

Impact on trials:

- Consent forms must be clear, concise, and understandable.
- Modern approaches like eConsent can enhance comprehension but must be validated.

- Special protections apply for minors, vulnerable populations, and emergency trials.

Who should pay attention in Principle 2 of the 11 Principles of ICH:

- **Sponsors:** Design Informed Consent Forms (ICFs) that are clear, translated, and aligned with regulatory and IRB/IEC expectations.
- **CROs:** Monitor informed consent processes and ensure documentation is complete.
- **Sites/SMOs:** Carry out the consent process, explain the trial to participants, and document all steps properly.
- **Labs:** Must ensure biological materials are used only within the scope of consent provided by participants.

2.3 Principle 3

“Clinical Trials should be subject to an independent review by an Institutional Review Board/Independent Ethics Committee (IRB/IEC).”

What it means:

- No clinical trial can begin without IRB/IEC approval. Trials must be conducted strictly according to the protocol approved by the IRB/IEC.
- Periodic review of the trial by the IRB/IEC is required during its conduct.

Impact on trials:

- Ethical and scientific review prevents poorly designed or unsafe studies.
- Ensures trials align with international ethical standards (Declaration of Helsinki, the Council for International Organizations of Medical Sciences (CIOMS)).
- Creates accountability through independent oversight.

Who should pay attention in Principle 3 of the 11 Principles of ICH:

- **Sponsors:** Prepare and submit all required documents for ethics approval; ensure amendments are also reviewed.
- **CROs:** Support submission management and track approvals across multiple countries.
- **Sites/SMOs:** Maintain communication with local IRBs/IECs and comply with conditions.
- **Labs:** Provide data or validation methods if their work is included in IRB/IEC submissions.

2.4 Principle 4

“Clinical Trials should be scientifically sound for their intended purpose and based on robust and current scientific knowledge and approaches.”

What it means:

- Available nonclinical and clinical information must justify the trial.
- Protocols must reflect the state of knowledge on the investigational product, the condition, and the intended population.
- Trials should be periodically reviewed in light of new data.

Impact on trials:

- Ensures valid results and protects participants by avoiding scientifically unsound research.
- Reduces wasted resources and ethical concerns around unjustified trials.

Who should pay attention in Principle 4 of the 11 Principles of ICH:

- **Sponsors:** Lead trial design with strong scientific rationale and periodically reassess ongoing trials.
- **CROs:** Provide statistical, methodological, and feasibility input.
- **Sites/SMOs:** Contribute to feasibility, ensuring the trial design is practical for the target population.
- **Labs:** Validate assays and confirm their suitability for reliable trial outcomes.

2.5 Principle 5

“Clinical Trials should be designed and conducted by qualified individuals.”

What it means:

- Trials require a multidisciplinary team of healthcare professionals, scientists, statisticians, ethicists, coordinators, auditors, and monitors.
- All must be qualified by education, training, and experience.

Impact on trials:

- Prevents protocol errors, safety issues, and data integrity problems caused by unqualified staff.
- Ensures ethical and scientific credibility.

Who should pay attention in Principle 5 of the 11 Principles of ICH:

- **Sponsors:** Confirm vendor and investigator qualifications during selection.

- **CROs:** Ensure staff training and maintain qualification records.
- **Sites/SMOs:** Keep CVs, training logs, and delegation logs up to date.
- **Labs:** Employ qualified personnel and document training in GCP and technical procedures.

2.6 Principle 6

“Quality should be built into the scientific and operational design and conduct of Clinical Trials.”

What it means:

- Quality is defined as “fitness for purpose”.
- Critical-to-quality factors (CTQs) must be identified prospectively.
- Strategies should detect and prevent serious noncompliance with GCP, the protocol, or regulations.

Impact on trials:

- Promotes Quality by Design (QbD), reducing protocol amendments and major findings in inspections.
- Enhances efficiency by focusing resources on the most critical elements.

Quality by Design (QbD) – an approach that builds quality into the study from the start by identifying critical factors that affect participant safety and data reliability, then designing processes to focus on those factors. It helps ensure trials are efficient, feasible, and more likely to meet their objectives.

Who should pay attention in Principle 6 of the 11 Principles of ICH:

- **Sponsors:** Build CTQs into trial design and establish robust QMS.
- **CROs:** Implement RBQM strategies and monitor CTQs throughout trial conduct.
- **Sites/SMOs:** Execute processes that protect participants and ensure critical data is reliable.
- **Labs:** Ensure laboratory processes are validated and proportionate to trial CTQs.

2.7 Principle 7

“Clinical Trial processes, measures and approaches should be implemented in a way that is proportionate to the risks to participants and to the importance of the data collected and that avoids unnecessary burden on participants and investigators.”

2.8 Principle 8

“Clinical Trials should be described in a clear, concise, scientifically sound and operationally feasible protocol.”

2.9 Principle 9

“Clinical Trials should generate reliable results.”

2.10 Principle 10

“Roles and responsibilities in Clinical Trials should be clear and documented appropriately.”

2.11 Principle 11

“Investigational Products used in a Clinical Trial should be manufactured in accordance with applicable Good Manufacturing Practice (GMP) standards and be managed in accordance with the product specifications and the trial Protocol.”

3. Conclusion

The 11 principles of ICH GCP E6(R3) provide a clear, modernized framework for conducting clinical trials. They place ethics and participant well-being at the centre, require scientifically robust protocols, embed quality and risk-based approaches into trial conduct, and link GCP more closely to GMP.

For biotech, pharma, CROs, sites, and labs, compliance is not just a regulatory obligation – it is essential for conducting credible, efficient, and **patient-centred** research. Early adoption of these principles helps organizations build trust with participants, investigators, and regulators alike.